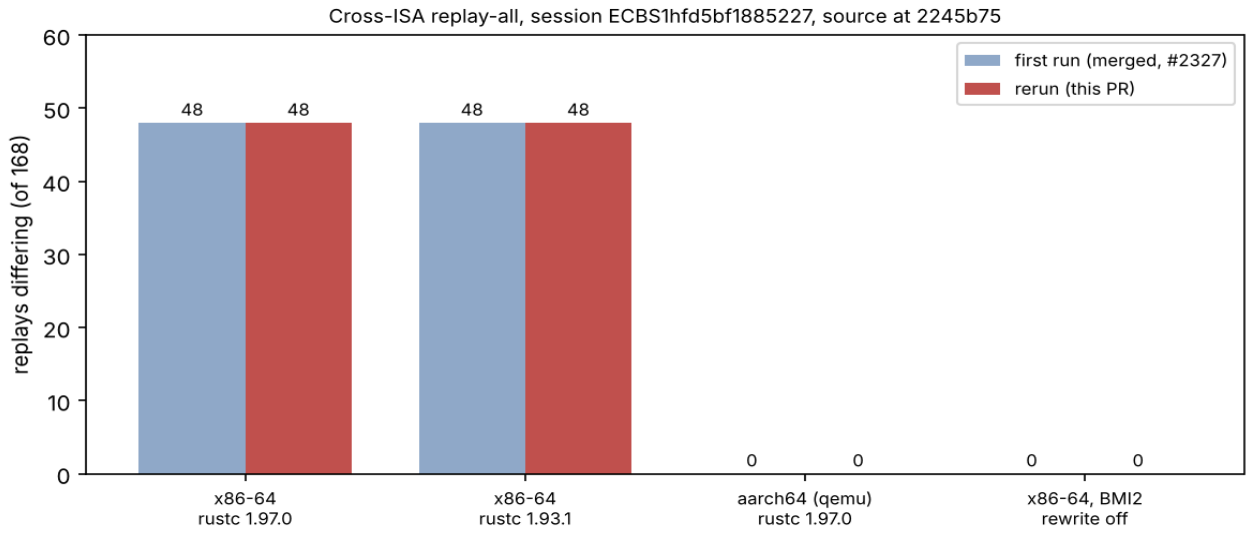
Item 1 cross-ISA experiment: independent rerun (2026-10-20)

Replication of `../item1_cross_isa_20261020/RESULT.md`, requested by the owner. Same method, fresh container, fresh toolchains and fresh builds of the recorded commit `2245b75`; nothing reused from the first run except the committed receipts it is compared to.



Result

The first run reproduces exactly. For all four builds the rerun's failing set is identical to the committed receipt, compared per sequence number over all 168 replays and per problem string.

build	first run	rerun	per-seq outcome identical
x86-64, rustc 1.97.0	48 differ	48 differ	yes (168/168 positions)
x86-64, rustc 1.93.1	48 differ	48 differ	yes
aarch64 under qemu-user, rustc 1.97.0	0	0	yes
x86-64, one-line patch: BMI2 word-wise rewrite off	0	0	yes

So the conclusion of the first run stands: the 48 mismatches are produced by the x86-64-only BMI2 word-wise rewrite in `inherited_f4.rs`; with it off, an x86 build reproduces the Apple M4 Pro counts on all 168 replays, as the aarch64 build does.

What is and is not independent

- Independent: container, rustup toolchains (re-downloaded), qemu-user and cross gcc (re-installed), builds, audits. The rerun was made by the same agent that ran the first, so it is a replication, not a second author's check.
- Not independent: the same recorded session, the same x86 host class (a cloud VM; the receipts' env-class ids show it), and the same method of switching the BMI2 branch off (one line, patch). The aarch64 build still runs under emulation, not on hardware.
- Binary hashes differ from the first run (`logs/binary_sha256.txt` against the first run's) because the build directory is embedded in the binaries (`/dev/shm/y` against `/dev/shm/x`); the audit outcomes do not.
- Not repeated: the "five files off" and "only `inherited_f4.rs` x86 paths off" builds (the one-line patch subsumes them) and the invalid scatter-env run.
- Counted-work reproducibility, not a speed claim: no `benchmarks/` entry, no leaderboard change. The session directory was only read.

Receipts: `receipts/audit_*.json`; logs: `logs/`.